Jack Lacy

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Education

University of Western Ontario — London, ON
MESc in Software Engineering – Vector Institute Specialization in AI — *Sep. 2025 – Aug. 2027*

University of Western Ontario — London, ON
BESc in Software Engineering with Co-op – Dean's Honour List — *Sep. 2020 – Apr. 2025*

Experience

Graduate Teaching Assistant — Sep. 2025 – Present
Western University – Databases (3rd year) & Software Engineering Capstone (4th year) — *London, ON*

- Lead weekly databases labs for 100+ students on design, SQL, and normalisation. Responsible for grading all assignments. Also proctored midterms and finals.
- Mentor software engineering capstone teams through a full-year project, advising on requirements, architecture, and development process, and evaluating milestone deliverables.

Software Development Intern — May. 2023 – Sep. 2024
CARFAX — *London, ON*

- **Document scraping engine:** Enhanced a Java and AWS engine with ML-based computer vision integration that extracts data from 5000+ records per day.
- **Template management tool:** Built a TypeScript/React tool that QA uses to onboard new forms, expanding the engine's support to 50+ new form types.
- **Deployment automation:** Replaced manual deployments with GitLab CI/CD pipelines and Terraform infrastructure as code for both projects.
- Worked on a 5-person Agile team, practicing pair programming and test-driven development (TDD).

Projects

AI-Assisted Dental Skills Assessment — May. 2026 – Present
MESc Thesis Research – Python, YOLO, LLMs, React Native, ONNX

- Developing a computer vision system for dental education that combines multiple YOLO models, each assessing a different aspect of students' cavity preparations, to deliver several forms of feedback.
- Integrating LLMs to turn model outputs into clear written feedback, delivered through a companion React Native app.

Robustness of Vision Language Models to Visual Prompt-Injection — Jan. 2026 – Apr. 2026
Presented at the IUW Conference, University of Waterloo

- Evaluated Qwen3-VL (4B, 8B, and 4-bit quantized) and Gemini 3.1 Flash Lite on 973 real measurement-reading images from MeasureBench, measuring accuracy, MAE, MAPE, and attack success rate (ASR).
- Found larger, unquantized models more vulnerable with Gemini reaching 86% ASR, dropping its accuracy from 29% to 4%, while defensive prompting cut Qwen3-VL-8B's ASR from 59% to 24%.

LLM Story-Point Estimation — Sep. 2025 – Dec. 2025
Research Project

- Fine-tuned Google's Gemma-3 in PyTorch into a local story-point estimator with an MAE under 1 point.

Western Building Navigation — Sep. 2024 – Apr. 2025
Capstone Project

- Led a team of 4 building indoor navigation to 100+ campus rooms placing 3rd place of 30+ capstone teams.

Toboggan Showcase Website — Sep. 2024 – Apr. 2025
Western Engineering Toboggan Team – Software Development Lead

- Built a Three.js 3D showcase site presented at the Great Northern Concrete Toboggan Race exhibition.

Technical Skills

Languages: Java, TypeScript, JavaScript, Python, C/C++, C#, SQL, Terraform
Frameworks: React, React Native, Node.js, ExpressJS, Spring, JUnit, Jest
AI/ML: PyTorch, YOLO, Qwen3-VL, Gemma-3, LLM fine-tuning and evaluation
Developer Tools: Git, Docker, AWS, GCP, GitLab CI/CD, Jenkins, Gradle | **Certified**: AWS Cloud Practitioner